

Management Information Systems

05. Building and Managing Information Systems

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Building and Managing Systems

- Organizational Change and Business Process Redesign
- Core Activities in the Systems Development Process
- Principal Methodologies and Principal Approaches to Systems Development
- Managing Projects



Organizational Change and Business Process Redesign



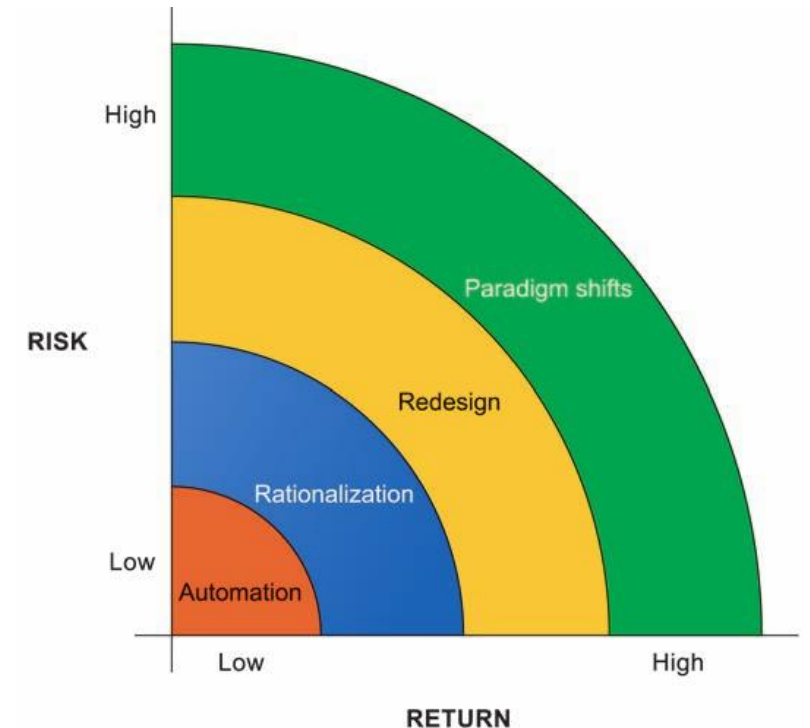
What are the First Steps in Building an Information System?

- Recall the Knopp (诺普集团) case?
- First steps: strategic planning
 - What are the business challenges to address?
 - What is the goal of the information systems?
 - What organizational changes and managerial moves are needed?

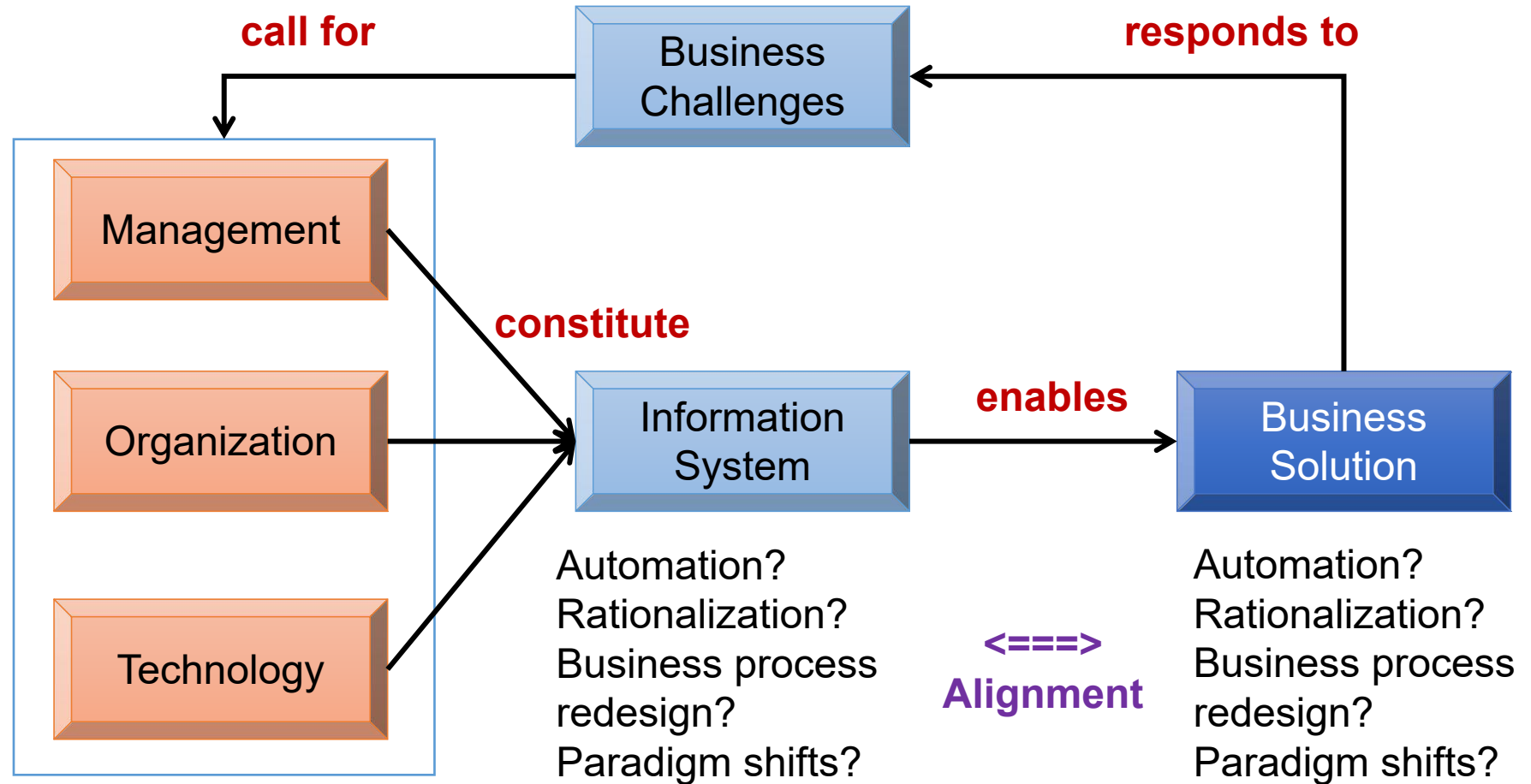


Systems Development and Organizational Change

- Automation (自动化)
 - Increases efficiency
 - Replaces manual tasks
- Rationalization (合理化)
 - Streamlines standard operating procedures
 - Often found in programs for making continuous quality improvements: e.g., Total quality management (TQM) and “Six sigma”
- Business process redesign (业务流程重组)
 - Analyze, simplify, and redesign business processes
 - Reorganize workflow, combine steps, eliminate repetition
- Paradigm shifts (模式转换)
 - Rethink nature of business
 - Define new business model
 - Change nature of organization



Alignment Is Critical

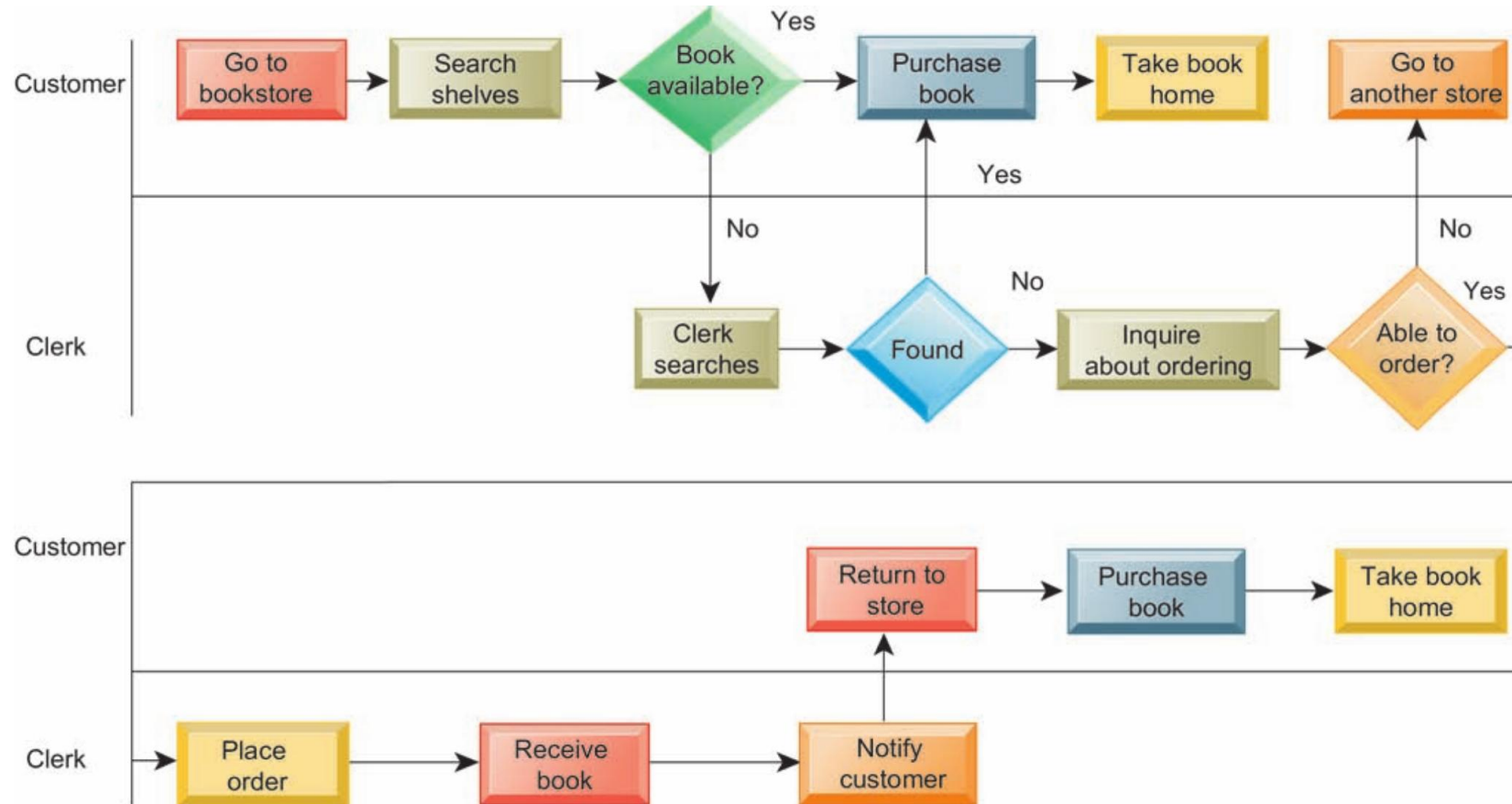


Business Process Redesign

- Also called: Business Process Re-engineering (BPR)
 - Analyze, simplify, and redesign business processes
 - Reorganize workflow, combine steps, eliminate repetitive, paper-intensive tasks.
- Business Process Management (BPM)
 - Variety of tools, methodologies to analyze, design, optimize processes
 - Used by firms to manage business process redesign

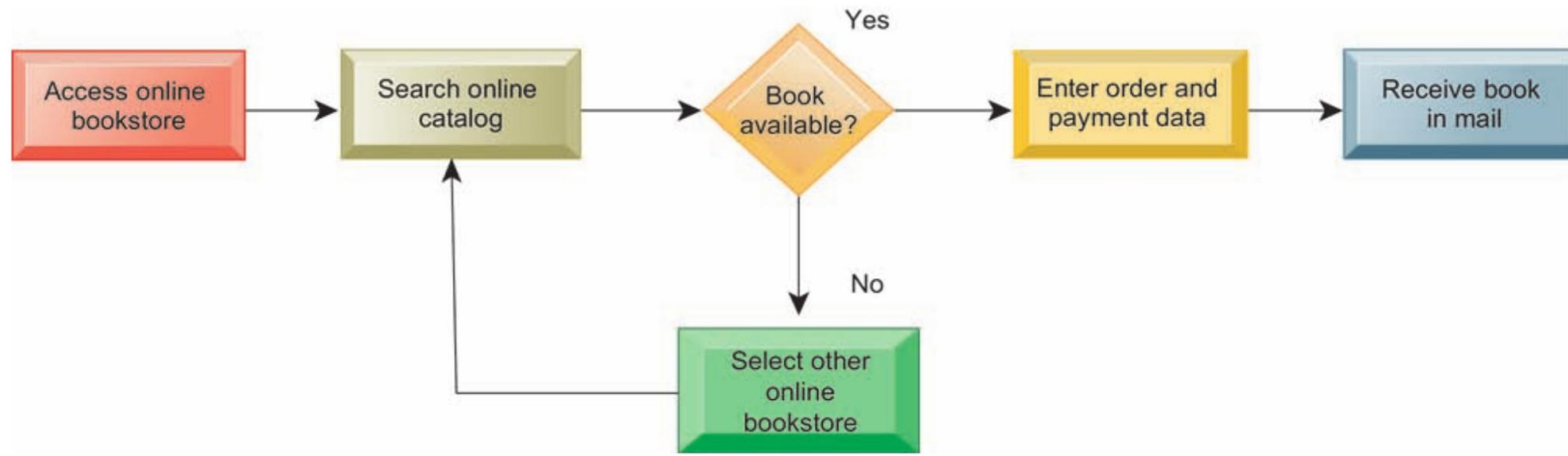


BPR Example



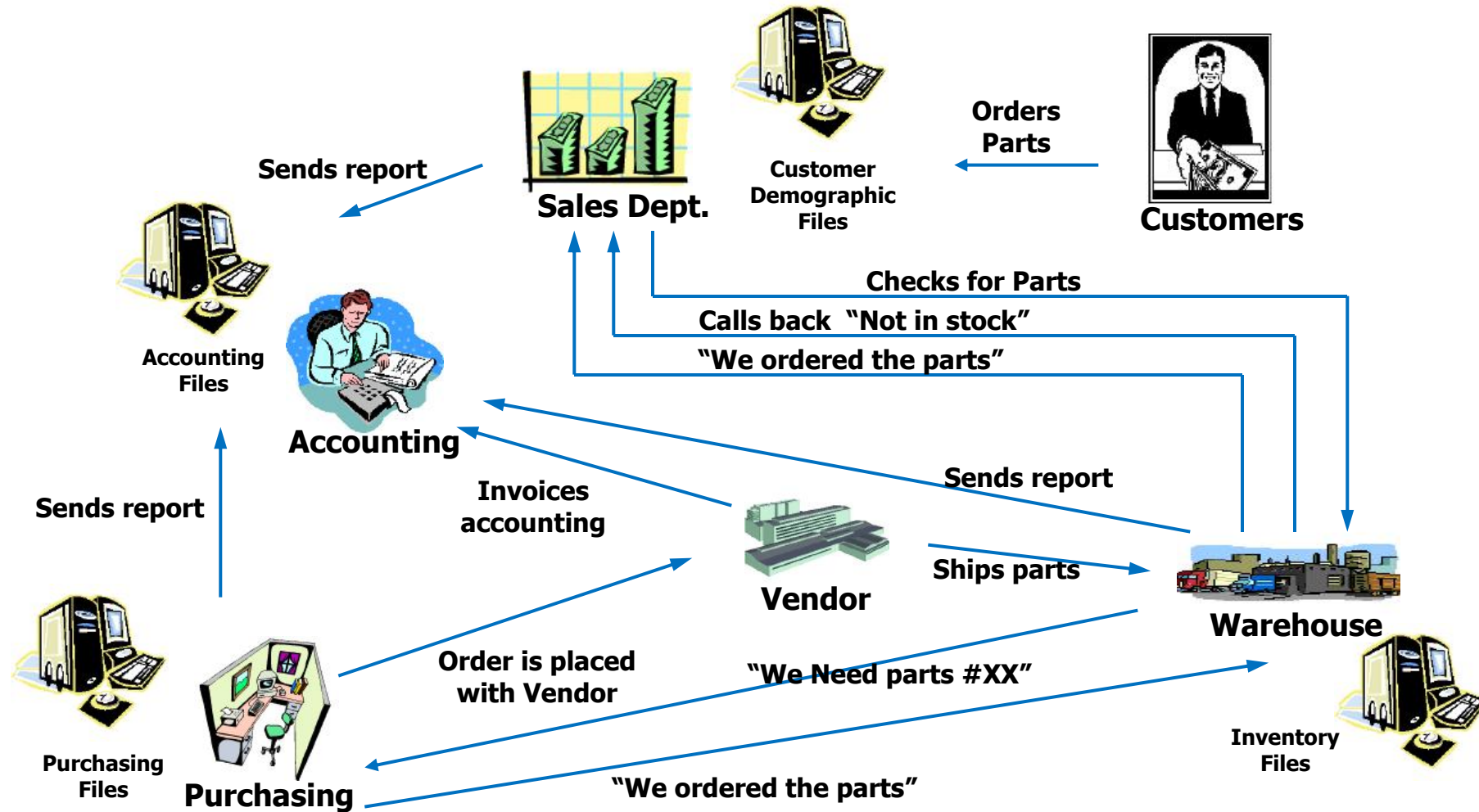
Purchasing a book from a physical bookstore requires many steps to be performed by both the seller and the customer.

BPR Example



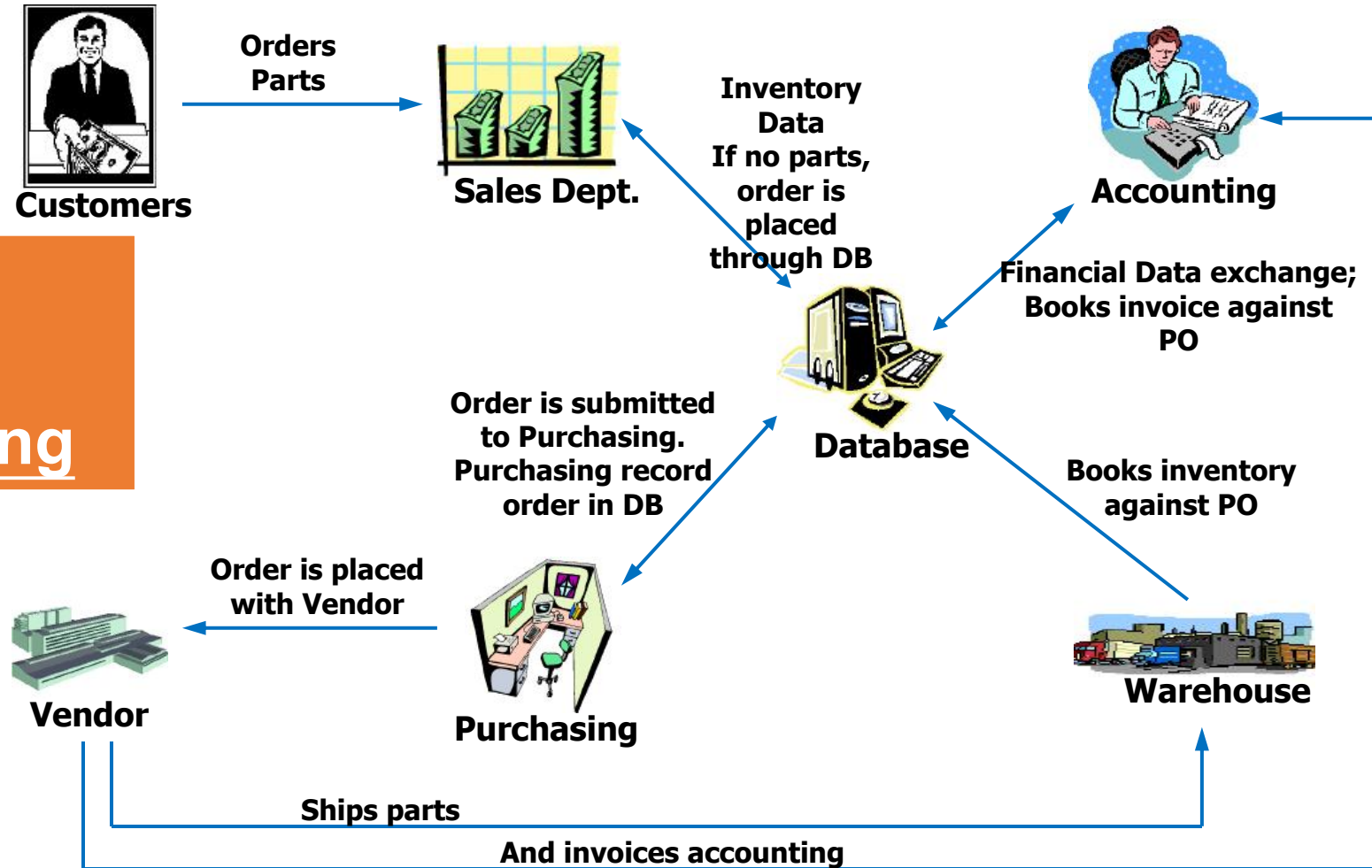
Using Internet technology makes it possible to redesign the process for purchasing a book so that it requires fewer steps and consumes fewer resources.

Business Process Before Implementing ERP



Business Process after Implementing ERP

Business Process Reengineering



Steps of BPR/BPM

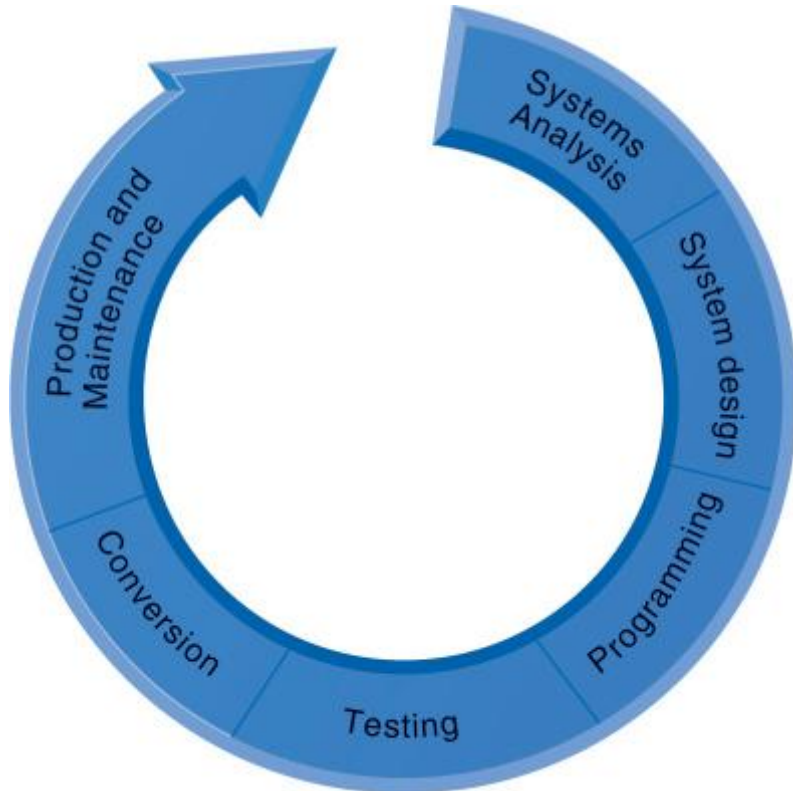
- Identify processes for change
- Analyze existing processes
- Design the new process
- Implement the new process
- Continuous measurement



Core Activities in the Systems Development Process



Systems development Activities



CORE ACTIVITY	DESCRIPTION
Systems analysis	Identify problem(s) Specify solutions Establish information requirements
Systems design	Create design specifications
Programming	Translate design specifications into program code
Testing	Perform unit testing Perform systems testing Perform acceptance testing
Conversion	Plan conversion Prepare documentation Train users and technical staff
Production and maintenance	Operate the system Evaluate the system Modify the system

Systems Analysis

- Analysis of problem to be solved by new system
 - Defining the problem
 - Identifying causes
 - Process modeling (**as-is**)
- Specifying solutions
 - Identifying information requirements
 - Process modeling (**to-be**)
- Feasibility study
- Systems proposal report



Systems Design

- Describes system specifications that will deliver functions identified during systems analysis
- Should address all managerial, organizational, and technological components of system solution
- Role of end users
 - User information requirements drive system building
 - Users must have sufficient control over design process to ensure system reflects their business priorities and information needs
 - Insufficient user involvement in design effort is major cause of system failure



Programming and Testing

- Programming
 - System specifications from design stage are translated into software program code
- Testing
 - Ensures system produces right results
 - Unit testing: Tests each program in system separately
 - System testing: Test functioning of system as a whole
 - Acceptance testing: Makes sure system is ready to be used in production setting
 - Test plan: All preparations for series of tests



Conversion (切换)

- Conversion
 - Process of changing from old system to new system
- Four main strategies
 - Parallel strategy
 - Direct cutover
 - Pilot study
 - Phased approach
- Requires end-user training
- Finalization of detailed documentation showing how system works from technical and end-user standpoint



Production and Maintenance

- System reviewed to determine if revisions needed
- May include post-implementation audit document
- Maintenance
 - Changes in hardware, software, documentation, or procedures to a production system to correct errors, meet new requirements, or improve processing efficiency
 - 20 percent debugging, emergency work
 - 20 percent changes to hardware, software, data, reporting
 - 60 percent of work: user enhancements, improving documentation, recoding for greater processing efficiency



Principal Methodologies and Approaches



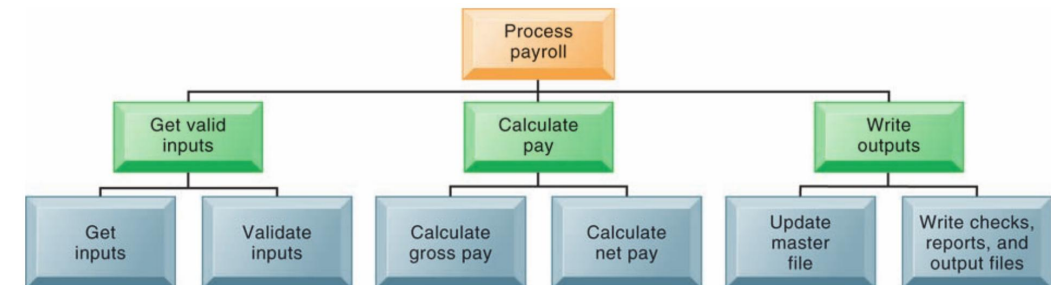
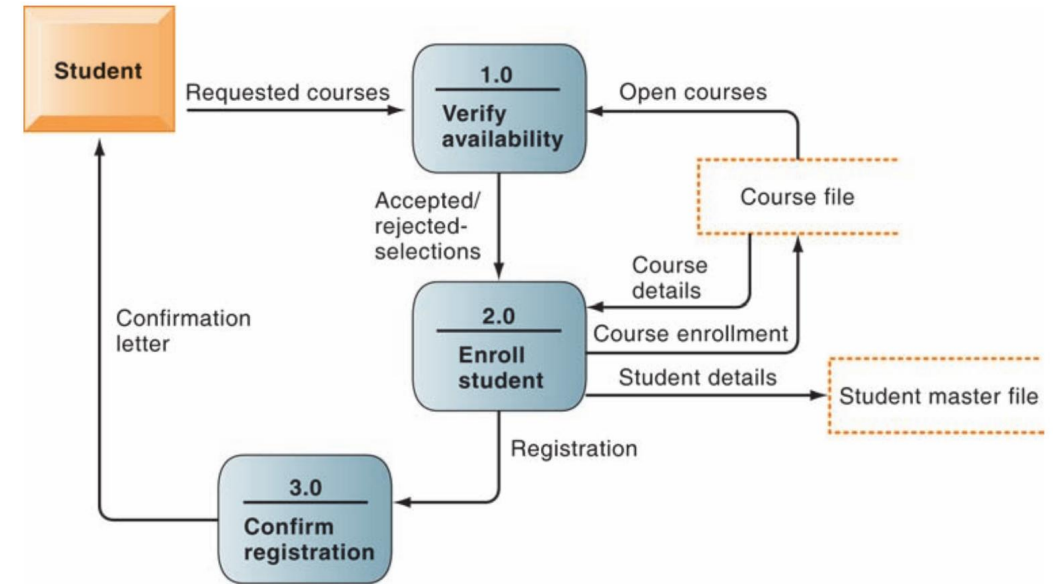
Methodologies and Approaches

- Methodologies
 - methods used for modeling and designing systems
 - used in analysis, design, programming, and testing activities
 - two major alternatives: (1) structured methodologies; (2) object-oriented development
- Approaches
 - methods used for organizing the core activities
 - major alternatives: (1) the traditional systems life cycle; (2) prototyping; (3) application software packages; (4) end-user development; (5) outsourcing



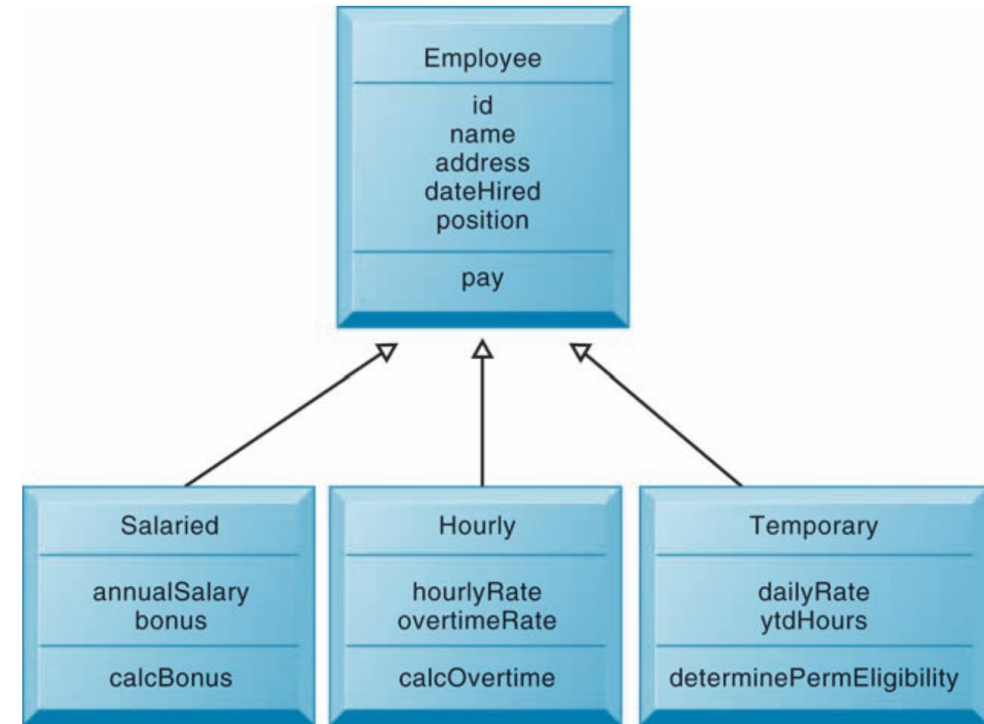
Structured Methodologies

- Structured: Techniques are step-by-step, progressive
 - Process-oriented: Focusing on modeling processes or actions that manipulate data
- Separate data from processes
- Typical tools
 - ER diagram (ERD): represent the data structure
 - Data flow diagram (DFD): represents system component processes and flow of data between them
 - Structure chart: top-down chart, showing each level of design, relationship to other levels, and place in overall design structure



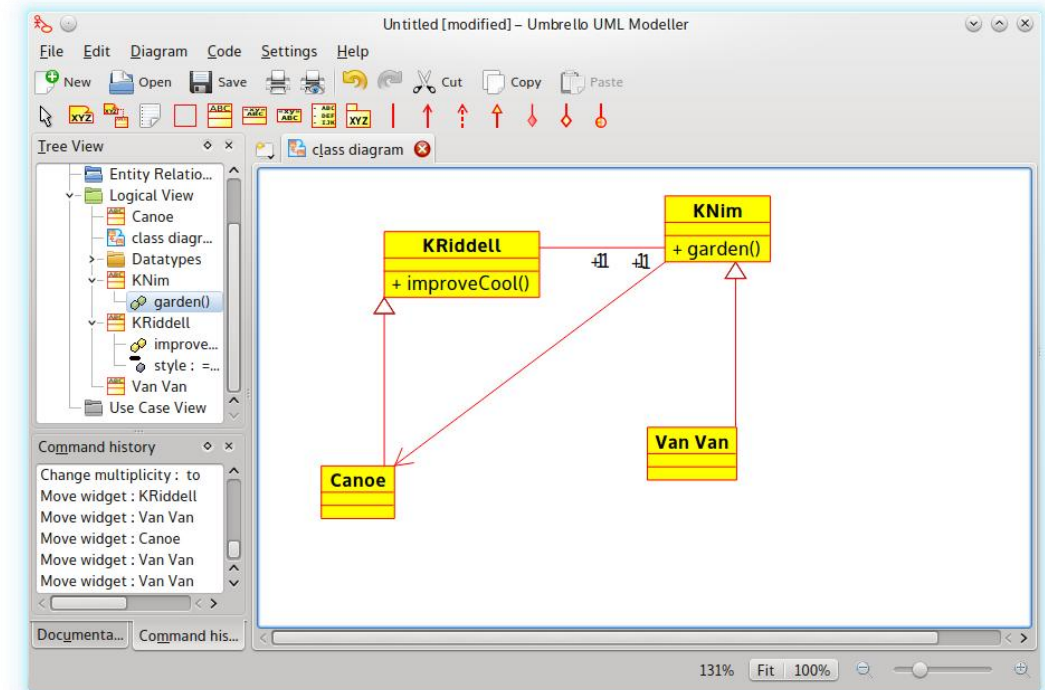
Object-Oriented Development

- Object
 - Basic unit of systems analysis and design
 - Combines data and the processes that operate on those data
 - Data in object can be accessed only by operations associated with that object
- Object-oriented modeling
 - Based on concepts of class and inheritance
 - Objects belong to a certain class and have features of that class
 - May inherit structures and behaviors of a more general, ancestor class
- More iterative and incremental
- Potentially reduce time and cost: objects are reusable



Computer-Aided Software Engineering (CASE)

- CASE: software tools to automate development and reduce repetitive work, including:
 - Graphics facilities for producing charts and diagrams
 - Screen and report generators, reporting facilities
 - Analysis and checking tools
 - Data dictionaries
 - Code and documentation generators
- Widely accepted standard: UML



CASE tool example: Umbrello

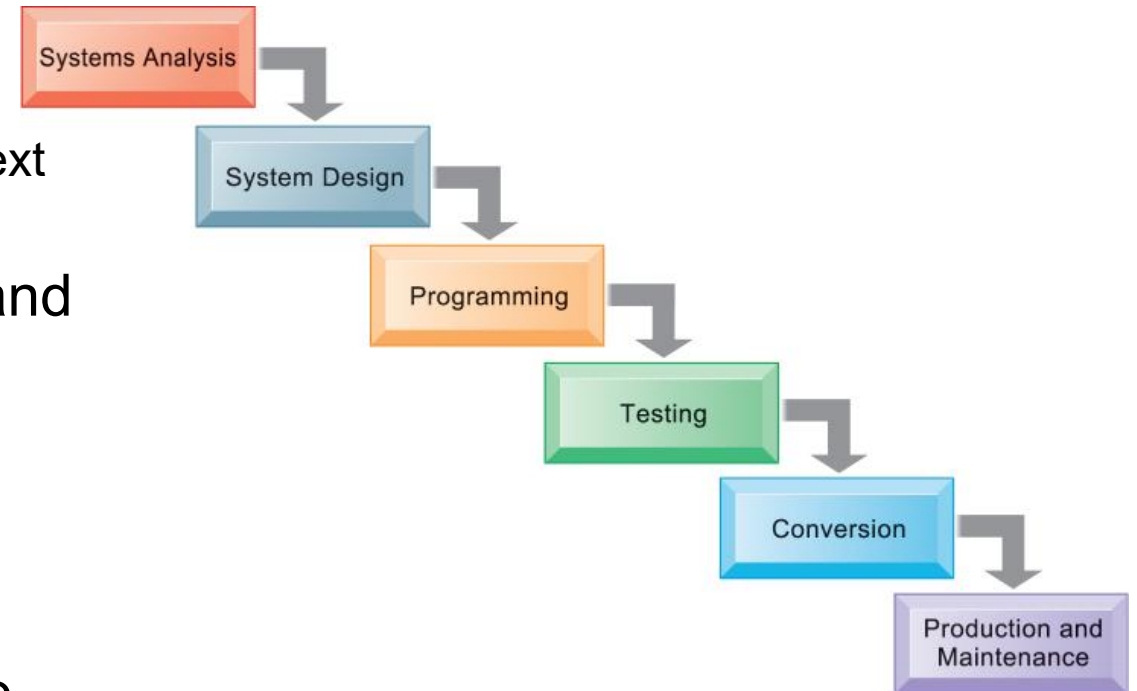
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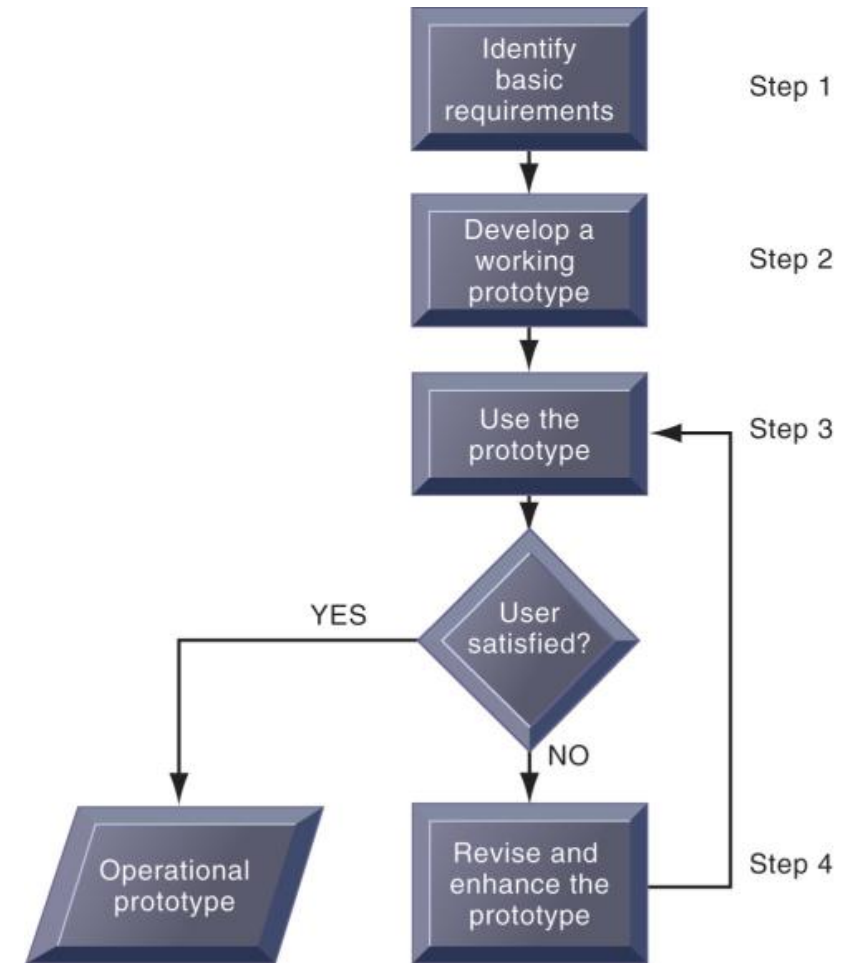
Traditional Systems Life Cycle (生命周期法)

- Oldest method for building information systems
- Phased approach
 - Development divided into formal stages
 - “Waterfall” approach: One stage finishes before next stage begins
- Formal division of labor between end users and information systems specialists
- Emphasizes formal specifications and paperwork
- Still used for building large complex systems
- Can be costly, time-consuming, and inflexible



Prototyping (原型法)

- Building experimental system rapidly and inexpensively for end users to evaluate
- Prototype: Working but preliminary version of information system
 - Approved prototype serves as template for final system
- Advantages of prototyping
 - Useful if some uncertainty in requirements or design solutions
 - Often used for end-user interface design
 - More likely to fulfill end-user requirements
- Disadvantages
 - May gloss over essential steps
 - May not accommodate large quantities of data or large number of users
 - May not undergo full testing or documentation



End-User Development

- Allows end users to develop simple information systems with little or no help from technical specialists
- Reduces time and steps required to produce finished application
- Advantages
 - More rapid completion of projects
 - High level of user involvement and satisfaction
- Disadvantages
 - Not designed for processing-intensive applications
 - Inadequate management and control, testing, documentation
 - Loss of control over data
- **Could be increasingly effective with the support of AI**



Application Software Packages and Cloud Software Services

- Application software packages and cloud software services
 - Save time and money
 - Many packages offer customization features
- Evaluation criteria for systems analysis include:
 - functions provided
 - flexibility
 - user friendliness
 - vendor quality
 - required resources
 - organizational change



Outsourcing (外包)

- Forms
 - Cloud and SaaS providers
 - External vendors to design, create software
- Domestic outsourcing
 - mostly driven by firm's need for additional skills, resources, assets
- Offshore outsourcing
 - mostly driven by cost-savings
- Risks
 - Hidden costs, e.g., identifying and selecting vendor, coping with cultural differences
 - Opening up proprietary business processes to third party



Managing Projects

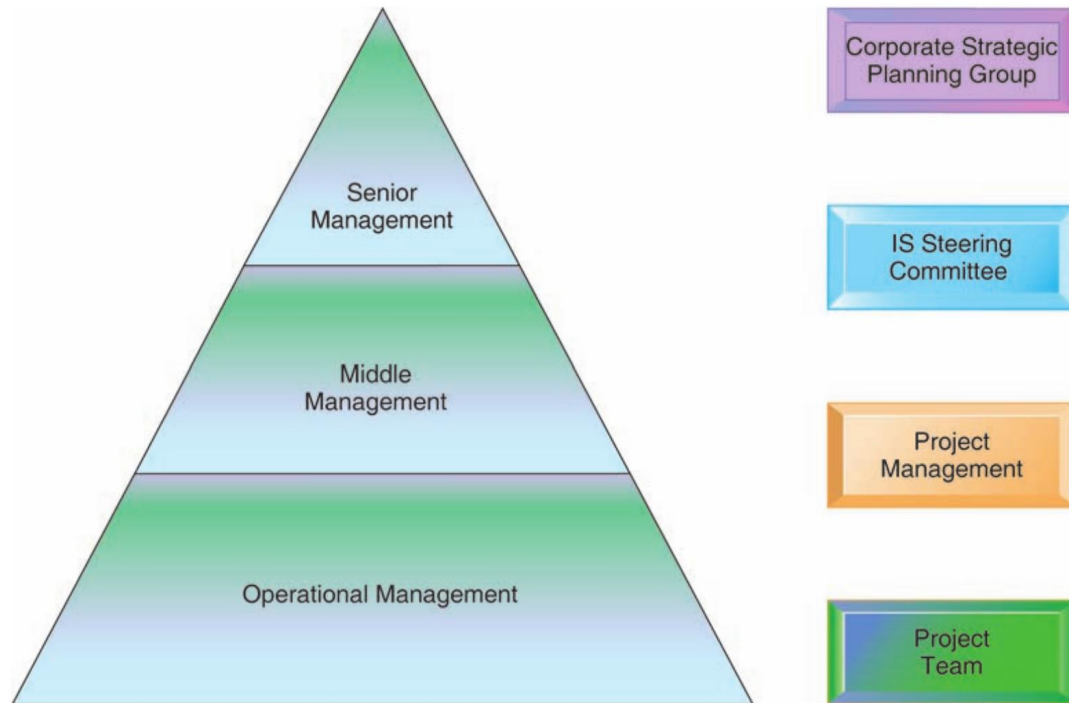


Managing Projects

- Selecting and Evaluating Information Systems Projects
- Costs and Benefits Analysis
- Risks Management



Management Structure for Information Systems Projects



- Responsible for firm's strategic plan
- Reviews and approves plans for systems in all divisions
- Responsible for overseeing specific projects
- Responsible for individual systems project

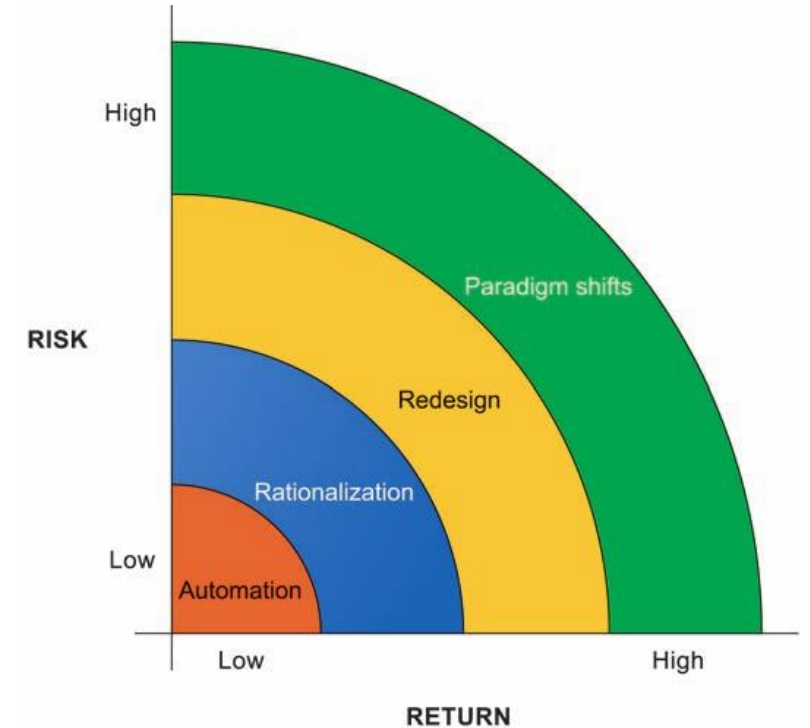
Linking Systems Projects to The Business Plan

- First of all, define the target (remember the “Knopp” case?)
 - identify the information systems projects that will deliver the most business value
- Information systems plan (信息系统规划书)
 - supports overall business plan
 - strategic systems are incorporated into top-level planning
- Typical structure of an information systems plan
 - Purpose of plan
 - Strategic business plan rationale
 - Current systems/situation
 - New developments
 - Management strategy
 - Implementation plan
 - Budget requirements



Understanding the Nature of the Project

- Automation (自动化)
 - Increases efficiency
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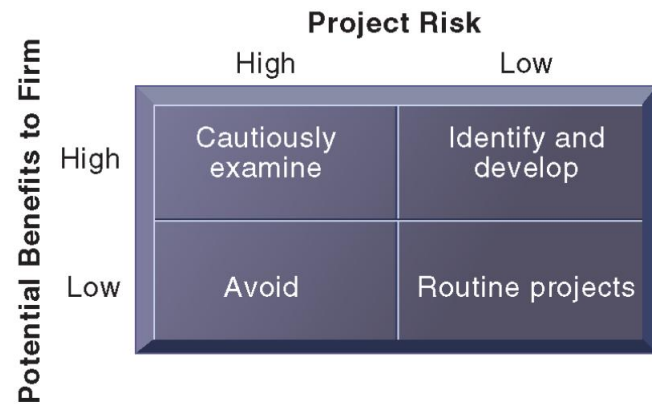


Information Systems Plan

- 1. Purpose of the Plan
 - Overview of plan contents, Current business organization and future organization, Key business processes, Management strategy
- 2. Strategic Business Plan Rationale
 - Current situation, Current business organization, Changing environments, Major goals of the business plan, Firm's strategic plan
- 3. Current Systems
 - Major systems supporting business functions and processes, Current infrastructure capabilities, Difficulties meeting business requirements, Anticipated future demands
- 4. New Developments
 - New systems projects (Project descriptions, Business rationale, Applications'role in strategy), New infrastructure capabilities required,
- 5. Management Strategy
 - Acquisition plans, Milestones and timing, Organizational realignment, Internal reorganization, Management controls, Major training initiatives, Personnel strategy
- 6. Implementation Plan
 - Anticipated difficulties in implementation, Progress reports
- 7. Budget Requirements
 - Requirements, Potential savings, Financing, Acquisition cycle



Considering New Developments: Portfolio Analysis



- Used to evaluate alternative system projects
- Inventories all of the organization's information systems projects and assets
- To improve return on portfolio, balance risk and return from systems investments
 - there is no ideal profile for all firms
 - information-intensive industries (e.g., finance) may have a few high-risk, high-benefit projects to ensure that they stay current with technology
 - non-information-intensive industries should focus on high-benefit, low-risk projects

A Real-world Case

- A large investment holding group company (大型投资控股集团公司)
 - Highly diversified business scope: chemical raw materials, agricultural products, daily necessities, electronic products,...
 - Numerous independently operating subsidiaries
 - Extensive capital operations
- IT planning
 - How to improve the competitiveness of the group through **ERP implementation**?
- Project choices
 - (1) an ERP system
 - (2) a CRM system
 - (3) a knowledge management system
 - (5) ESS and DSS
 - (6) integrated financial reporting system



Another Real-World Case

- A private enterprise mainly engaged in public utilities (gas, etc.)
 - Stable and mature business in Hebei
 - At the moment, the nationwide utility sector was just opening up to private companies
 - Seeking to seize the opportunity and expand its markets
- IT planning
 - How to use ERP to achieve rapid growth?
- Project choices
 - (1) an ERP system
 - (2) a CRM system
 - (3) a TPS system
 - (4) a knowledge management system
 - (5) ESS and DSS
 - (6) integrated financial reporting system



Yet another Case

- A large agricultural industry group (大型农业产业集团)
 - Scale: tens of thousands of employees, dozens of state-owned and holding subsidiaries, dozens of joint venture companies
 - Business area: agriculture, food processing and delivery
 - Mission: to serve the rural areas and to provide green and healthy food for the city
- IT Planning
 - Warehouse and logistics management systems
 - Enterprise resource planning
 - Supply chain planning systems
 - B2B e-commerce platform



Costs and Benefits Analysis

- Costs
 - Hardware, Telecommunications, Software, Services, Personnel
 - Organizational disruptions
- Tangible benefits
 - Can be quantified and assigned monetary value
 - Increased productivity, lower operational costs, reduced workforce, ...
 - Transaction processing systems usually produce more measurable, tangible benefits
- Intangible benefits
 - Cannot be immediately quantified but may lead to quantifiable gains in the long run
 - More efficient customer service, enhanced decision making, ...
 - MIS, DSS, ESS produce more intangible benefits



Capital Budgeting for Information Systems

- Capital budgeting models
 - Measure value of long-term capital investment projects
 - Rely on measures of the firm's cash outflows and inflows
- Principle capital budgeting models used to evaluate IT projects
 - Payback method (回收期法)
 - Accounting rate of return on investment (ROI, 投资回报率)
 - Net present value (净现值)
 - Internal rate of return (IRR, 内部收益率)
- Limitations of financial models
 - focus on the financial and technical aspects, overlook the social and organizational dimensions
 - costs not considered: organizational disruptions
 - benefits not considered: intangible benefits such as more timely decisions



Dimensions of Project Risks

- Project size
 - Costs
 - Time
 - Number of organizational units affected
 - Organizational complexity
- Project structure
 - Structured, defined requirements run lower risk
- Experience with technology
 - Team familiar with hardware and software



Change Management

- Change management
 - to address the process of organizational change surrounding system building
- Challenging: very high failure rate (up to 70 percent for BPR projects)
 - Employee concerns about change
 - Resistance by key managers
 - Changing job functions, career paths, recruitment practices
- User-designer communication gap
 - **Information systems specialists** often have a highly technical orientation to problem solving, looking for **elegant and sophisticated technical solutions**
 - **Users** prefer systems that are oriented toward **solving business problems** or **facilitating organizational tasks**
- Change agent (变革代理人)
 - One role of systems analyst
 - Redefines the configurations, interactions, job activities, and power relationships of organizational groups



User-designer Communication Gap



USER CONCERNS

Will the system deliver the information we need for our work?

Can we access the data on our iPhones, BlackBerrys, tablets, and PCs?

What new procedures do we need to enter data into the system?

How will the operation of the system change employees' daily routines?

DESIGNER CONCERNS

What demands will this system put on our servers?

What kind of programming demands will this place on our group?

Where will the data be stored? What's the most efficient way to store them?

What technologies should we use to secure the data?

Controlling Risk Factors

- Managing technical complexity
 - Internal integration tools
- Managing large projects
 - Formal planning and control tools
 - e.g., Gantt charts and PERT charts
- Increasing User Involvement and Overcoming User Resistance
 - External integration tools



Managing Technical Complexity

- Internal integration tools
 - Project leaders with technical and administrative experience
 - Highly experienced team members
 - Frequent team meetings
 - Securing of technical experience outside firm if necessary



Managing Large Projects

- Formal Planning and Control Tools
 - Used for documenting and monitoring project plans
 - Help identify bottlenecks and impact of problems
- Gantt charts
 - Visual representation of timing and duration of tasks
 - Human resource requirements of tasks
- PERT charts
 - Graphically depict tasks and interrelationships
 - Indicate sequence of tasks necessary



Increasing User Involvement and Overcoming User Resistance

- External integration tools
 - Link work of implementation team to users at all levels
 - Filling the “user-designer communication gap”
- Strategies to overcome user resistance
 - User participation, education and training
 - Management edicts and policies
 - Incentives for cooperation
 - Improvement of end-user interface
 - Resolution of organizational problems prior to introduction of new system



Building and Managing Systems

- Organizational Change and Business Process Redesign
- Core Activities in the Systems Development Process
- Principal Methodologies and Principal Approaches to Systems Development
- Managing Projects



Part I:

Concepts and Principles of Information Systems

- What we have learned
 - Concepts, dimensions, and types of information systems
 - IT infrastructure in business contexts
 - Types, characteristics, benefits, and challenges of enterprise applications
 - Characteristics of digital goods, digital markets, e-commerce models; types, characteristics, benefits, and challenges of knowledge management systems and decision making tools
 - Principles of building and managing information systems
- What are next?
 - Systems analysis and design



Tasks

- Reading
 - Textbook: Chapters 13-14
- Reviewing
 - The whole textbook and all the five sessions
- A quiz
 - Friday, April 3, 10:00am - Sunday, April 5, 10:00am
 - on Yuketang (<https://pro.yuketang.cn>)
- Get ready for “systems analysis and design”

